

# Connectivity

David Shamlin August 2026

# Connectivity

**Note:** This handout focuses on connecting peripherals with cables and wirelessly via Bluetooth. We will discuss connecting to the Internet further when we discuss what the Internet is and how it works in later sections.

Computers have the ability to connect to other pieces of hardware. You can connect your computer to

- The Internet
- Printers
- Pointing devices like mice, trackpads, and styluses (like an [Apple Pencil](#))
- Speakers, headphones, and earbuds (like [AirPods](#))
- Monitors
- Storage devices

*These kinds of devices are generically referred to as **peripherals**.*

There are two different ways to connect these kinds of devices to your computer: physically and wirelessly. Physical connections are made using cables that you plug into ports on your computer and the peripheral device. Wireless connections are made via Bluetooth or Wi-Fi. Physical connections are generally faster than wireless connections (although you may not notice a significant difference between the two with normal use.) Physical connections are also more stable/reliable than wireless connections.

Storage devices that you connect to computers are often referred to as “external storage” or “external drives”. The word “external” is used in this context to differentiate this kind of storage from the storage that is inside your computer; the storage that is inside your computer is sometimes referred to as “internal storage” or “internal drive”. Similarly, speakers that you can connect to your computer are also sometimes referred to as “external speakers” (also to differentiate them from the speaker(s) that are built in to your computer; the latter are sometimes referred to as “internal speakers”).

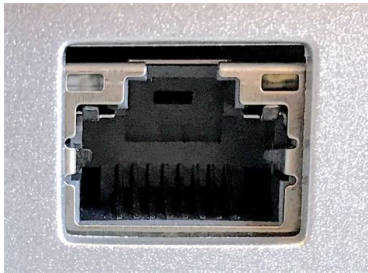
# Connectivity

Most of the kinds of devices you can connect to a laptop or desktop computer can be connected either physically or wirelessly while most of these kinds of devices only connect to phones and tables wirelessly. For some of these things, you can choose whether you want to connect them physically or wirelessly. Others can only be connected one way. Which ways you can connect depends on both the device you are connecting and the type of computer you are connecting it to. The table below shows the possible combinations of connections. There are some noteworthy differences. E.g., monitors can only be physically connected to desktops and laptops and can only be connected to phones and tables wirelessly.

|                       | Physical                                                                                                | Physical <b>or</b><br>Wireless                                                                                                                      | Wireless                                                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Desktops &<br>Laptops | <ul style="list-style-type: none"><li>• Monitors</li><li>• Storage Devices</li><li>• Speakers</li></ul> | <ul style="list-style-type: none"><li>• Internet</li><li>• Printers</li><li>• Mice</li><li>• Keyboards</li><li>• Headphones &amp; earbuds</li></ul> | <ul style="list-style-type: none"><li>• Styluses</li></ul>                                                                                             |
| Phones &<br>Tablets   | <ul style="list-style-type: none"><li>• Storage Devices</li></ul>                                       | <ul style="list-style-type: none"><li>• Headphones &amp; earbuds</li></ul>                                                                          | <ul style="list-style-type: none"><li>• Internet</li><li>• Printers</li><li>• Styluses</li><li>• Mice</li><li>• Keyboards</li><li>• Monitors</li></ul> |

# Physical connections: Types of Ports & Jacks

Computers have some “holes” on their cases where you plug in cables. These “holes” are called **ports**; the things you plug into ports are called **jacks**. Desktop and laptop computers tend to have several ports; smartphones and tablets have fewer. (The current generations of iPhones and iPads only have one port!). The images below show common types of ports.



An **ethernet** port (a.k.a. a LAN port); used to connect to the Internet. Only on desktop and laptop computers. Less commonly seen on laptops these day.



An **HDMI** port; used to connect to monitors. HDMI ports & cables send both video and audio to an attached monitor.



An **MagSafe** port; used to connect to a power source. To the best of my knowledge, MagSafe ports are only found on (some) Apple laptop and desktop computers.



An **lightning** port; found on some phones and tablets. Used to charge a device's internal battery and can be used to connect some peripherals. When used as the latter, it is acting as a USB port.

Note that a lightning port looks very similar to a USB-C port. A USB-C port has a set of pins in its center; a lightning port does not.



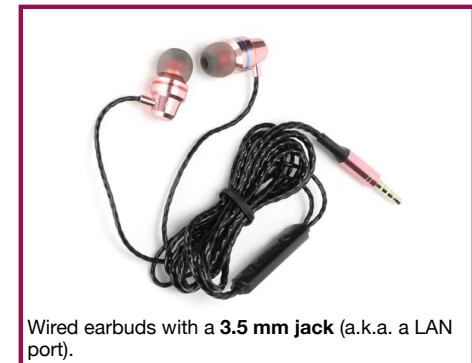
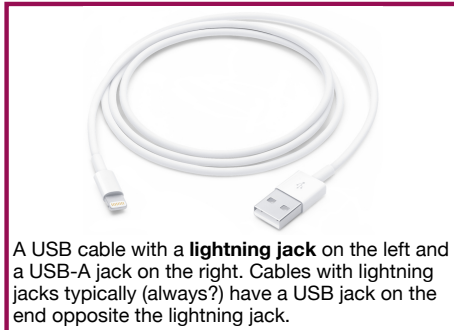
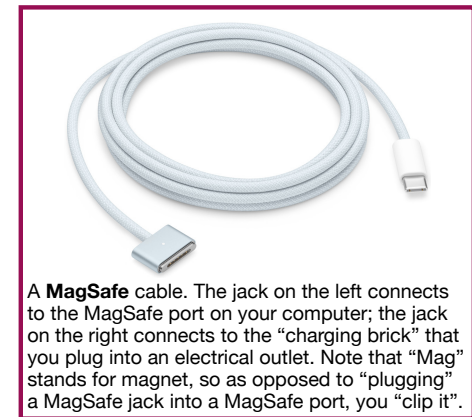
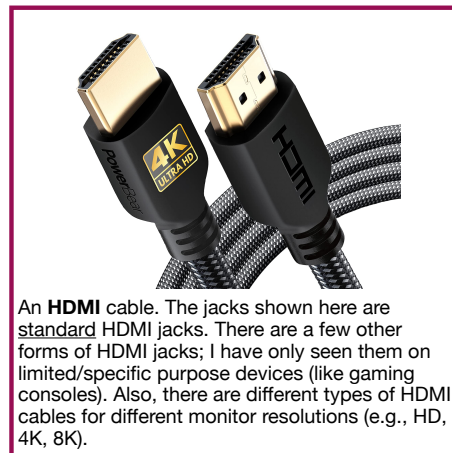
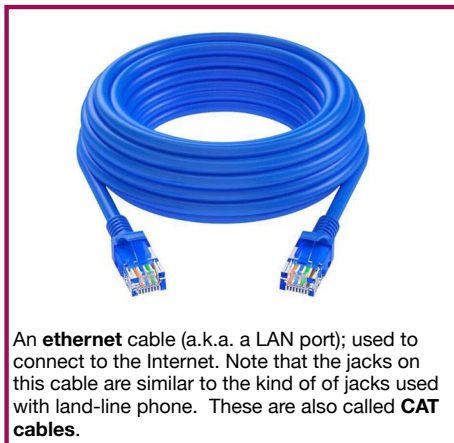
An **3.5 mm** port; used to connect speakers, headphones, and earbuds. I.e., an audio port.

These have become less commonly found on computers with the rise in popularity of wireless headphones/earbuds.



# Physical connections: Types of Ports and Jacks

The images below show examples of cables with the types of jacks that “match” the types of ports shown on the previous page.



# Physical connections: Types of Ports & Jacks

There quite a few “flavors” of USB ports and jacks. Type A, 3.0, and Type C appear on desktops, laptops, phones, and tablets. The other “flavors” are more likely to be found on peripherals and limited/special purpose devices (like Kindles/eReaders).

## USB CONNECTOR TYPES



All USB jacks/ports except Type C have an “up” and a “down”. When you plug one of these types of jacks into a matching type or port, you must orient the jack so that its “up” side matches the “up” side of the port.

The image below shows “head on” views of a Type A and/or 3.0 jacks and ports. Both the jack and the port in this image are correctly oriented to plug into each other.



# Physical connections: Adapters, Hubs, and Docks

When/if you need/want to physically connect peripherals to your computer, you may encounter a some challenges:

- The ports on your computer are of a different type than those on the peripheral or the jacks on the cable(s) you have on hand. In this type of situation, you can...
  - Use a cable that has different types of jacks on each end and match the types of ports your computer and peripheral that you are connect have. (See Figure 1.) E.g., my laptop has USB-C type ports and the external drive I use to backup my laptop has a USB 3.0 port.
  - Use an adapter that “converts” the jack at the end of a cable from one type to another. (See Figure 2.)
- Most laptops on the market today have two to three USB ports. If you have a laptop with only two USB ports, what do you do if you want to physically connect a mouse, keyboard, and external drive at the same time? Again, you have a couple of possible solutions:
  - Use a **hub**. (See Figure 3.) Hubs have a few USB ports and a single USB jack. You plug the jack into your computer and your peripherals into the ports on the hub. Hubs are typically physically small (i.e., fit in the palm of your hand) and designed to be easily portable.
  - Use a **dock**—a.k.a., a docking station. (See Figure 4.) Docks are designed to sit on a desk where you tend to frequently use your laptop; they are not designed to be as easily portable as hubs. They tend to have at least one of every kind of port. They also come with a power cord that you plug into a power outlet. They tend to have one USB Type 3 or 3.0 cable that you plug into your laptop. A dock makes it convenient to use with a laptop when you want to use several physically connected peripherals with your laptop at your desk. You connect all your peripherals and power to the dock and plug a single cable into your laptop. When you want to use your laptop “on the go”, you just have to unplug one cable and you’re good to go.



Figure 1



Figure 2



Figure 3



Figure 4

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  - Use an adapter that “converts” the jack at the end of a cable from one type to another.
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# Wireless connections: Wi-Fi, Cellular, & Bluetooth

Computers connect wirelessly to other devices using these three methods:

1. via Wi-Fi

2. via cellular networks

All kinds of smartphones support cellular network connections; some tablets do as well. No desktop or laptop computers available on the consumer market support cellular network connectivity.

Wi-Fi and cellular network connections share a similar primary purpose: accessing the Internet. Cellular connections additionally allow you to make phone calls. We will explore Wi-Fi and cellular networks in more detail in our section(s) on the Internet.

3. via Bluetooth

Bluetooth connections are used to connect wireless peripherals.

There is a fourth method sometimes used by peripherals like mice and keyboards. (Possibly only mice; I don't remember encountering any peripherals besides some mice that work this way.) Some wireless mice and keyboards come with a USB device similar to the image on the right. These are sometimes called "wireless dongle receivers". They work similarly to conventional Bluetooth wireless peripherals. Companies that make wireless peripherals that include dongle receivers do so to achieve a greater level of performance than they can achieve with the Bluetooth componentry that is built in to our computers.



There are a few "flavors" of wireless printers on the consumer market. Some only support Wi-Fi connectivity, some only support Bluetooth connectivity, some support both. Is one "flavor" superior to the others? Not necessarily; you may have a somewhat better experience with a Wi-Fi enabled printer in your home if there are other people living under your roof that will also use that printer. Wi-Fi printers connect to your home network; any person/computer can use it simply by connecting to your home network whereas every device you (or others in your home) want to use with a Bluetooth-only printer must explicitly **pair** each device to the printer. "Pairing" is a Bluetooth specific action you must take to use any Bluetooth enabled device with a given device.

# Disconnecting External Drives

If you use an external drive, there is a step you need to perform before you physically disconnect (i.e., unplug) it from your laptop or desktop computer. You need to **eject** it first.

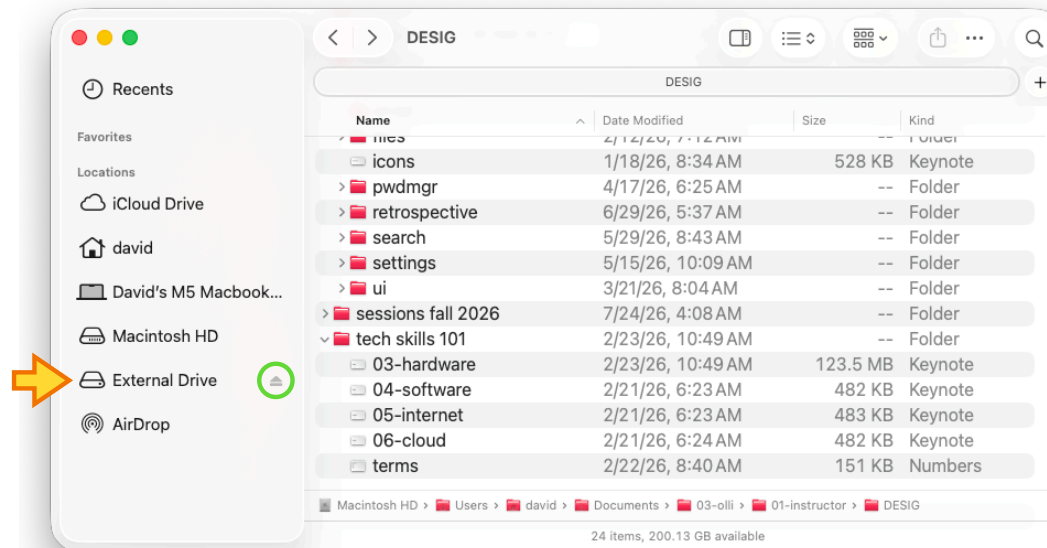
If you physically disconnect an external drive from your computer while your computer is actively reading/writing data to it, one of two bad things can potentially happen:

1. If your computer is in the process of writing data to a file on the external drive, some or all of that data will not get written to the external drive. This can happen if your external drive is either an HDD or a SSD. If this were to happen, you will have a problem when you reconnect your external drive at a later time and try to open the file. The file may not exist on the external drive or it may be incomplete (a.k.a. “corrupted”) in which case, the app you want to open the file with will not be able to read it.
2. If the external drive you are using is an HDD type of drive and you physically disconnect it while your computer is reading **or** writing data to the external drive, the drive could get physically damaged. When an external HDD is “in use” (i.e., the computer is reading data from/writing data to the drive), the platter is spinning and the arm is moving back and forth across the platter; if power (i.e., electricity) is “cut” cut from the drive (which happens as soon as you unplug it), the arm can “hit” (i.e., physically land on) the platter. This can physically damage the plate and/or the arm resulting in making the drive entirely unusable. I.e., you broke your drive.

Note that you **do not** need to eject external drives from iPhones and iPads.

# How to eject an external drive from a Mac

You use the Finder app (🔍) on your Mac to eject external drives. When you have an external drive connected to your Mac, you can see it in the sidebar of the Finder app. The screenshot below shows an external drive I connected to my Mac (yellow arrow); note external drives appear under the “Locations” section in the sidebar. External drives have names on Macs; I explicitly named my drive “External Drive” for this example. To the right of the drive’s name is a triangular shaped icon (green circle); the icon is the “eject button”. All I have to do to eject the external drive is click that icon. After a few seconds, the “row” in the sidebar will disappear. That’s the signal that the drive has been successfully ejected and it’s safe for me to physically disconnect my external drive from my laptop.



Reference: See the “Eject a storage service” section on [this page](#) in the Mac User Guide.

# Bluetooth Connections

Your computer has a bluetooth component (called a “bluetooth antenna”) inside it. When the antenna is “turned on”, it is constantly scanning the area around your computer for other bluetooth devices. (The “area” in this context is a sphere with about a 30’ radius.) When your computer detects a bluetooth peripheral it recognizes, your computer automatically connects to it; the bluetooth peripheral must be turned on for the connection to be made. The connection between your computer and the bluetooth peripheral will be maintained until (1) you turn off your computer or the bluetooth peripheral, (2) your computer “goes to sleep”, or (3) you physically move one of the two device too far away for the connection to be maintained (again about 30’ apart). If your devices lose their connection for any , they will normally automatically reconnect when you wake your computer up. If you your devices lose their connection for any of these reasons, they will normally automatically reconnect when you turn your computer back on, wake your computer up, or move the devices back within range of each other.

To enable your computer to recognize a bluetooth peripheral, you need “introduce” them to each other the first time you use the peripheral with your computer. The process of “introducing” a bluetooth peripheral to a computer is called **pairing**. When you purchase a bluetooth peripheral, it will come with some instructions on how to complete the pairing process. In my experience, a bluetooth device’s pairing process is something like this:

1. Turn on the bluetooth peripheral when you are near your computer. Your computer must also be turned on.
2. Put the peripheral in **pairing mode**. The peripheral manufacture’s instructions will tell you exactly how to do this. In my experience, it often involves pressing the peripheral devices on/off button some number of seconds.
3. In the Bluetooth control panel of your computer’s system settings, you will see the peripheral appear. There will either be a “pair” or “connect” button for your new peripheral in the control panel. Click/tap that button. After a few seconds, the control panel should then indicate that the peripheral is connected and you’re done!

Pairing some bluetooth peripherals may require an additional step: entering a 4-6 digit code in the peripheral. If you need to take this additional step, your computer will show you the code; you enter it into the peripheral device. This additional step mostly occurs when pairing a bluetooth keyboard to a computer and pairing a car with a smartphone. Bluetooth peripherals that include this step when they are paired for security reasons. I.e., it inhibits bad actors from pairing/using bluetooth devices to your computer and car without your awareness/consent.



# Turning wireless connectivity on and off

You can turn wireless connectivity on/off on your computer. To the best of my awareness, the ability to do this was introduced due to air travel regulations. I.e., for a period of time after wireless connectivity was first introduced in digital devices, air travel regulatory agencies (like the FAA) were concerned that use of wireless devices during flights would interfere with other digital systems on airplanes, so airlines required you to turn them off before planes kicked back from gates. Those days are behind us (although some airlines still request that you disable some wireless forms of connectivity during takeoffs and landings.)

The Wi-Fi and Bluetooth control panels in your computer's System Settings will have controls that allow you to turn both on and off.

On iPhones and iPads, you can swipe your finger from the upper right corner of the screen toward the center of the screen to reveal a special control panel that also allows you to do this. You will see an airplane icon on that panel that you can tap. When that icon is orange, Wi-Fi and Bluetooth are both turned off.

